MicroBioAgent: an AI agent for automated microbiome analysis

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Abstract

Microbiome studies require analytical choices that depend on the biological question, available assays, and intermediate findings. Translating these choices into executable analyses requires connecting heterogeneous methods, software environments, reference resources, and intermediate outputs. Here we present MicroBioAgent, a domain agent that translates biological questions into study-specific analyses across 16S amplicon, reference-based shotgun, and assembly-based metagenomic workflows. MicroBioAgent uses microbiome-specific skills to select and execute analytical methods in user-controlled computational environments. It evaluates intermediate results to determine subsequent analyses in line with the biological question. We evaluated MicroBioAgent on three published datasets through amplicon and shotgun analysis of tomato rhizosphere communities, taxonomic and functional profiling with classifier training in a colorectal cancer cohort, and metagenomic assembly and MAG generation from a model polyethylene degradation study, followed by taxonomic profiling and annotation of genes associated with alkane and oxidized-derivative degradation in focal MAGs. The reanalyses identified biological patterns consistent with the original studies, with differences in statistical support and taxon detection across analytical methods and reference databases. In robustness evaluations across 27 independent sessions spanning three language models and three levels of prompt detail, MicroBioAgent completed 98.8% of analytical tasks with consistent diversity and differential-abundance outputs. Ablation of the runtime patch showed that pausing the conversation turn during background execution reduced mean token consumption by more than half compared with repeated status checks, while enabling mid-run plan revision and automatic continuation upon task completion. These applications demonstrate that MicroBioAgent can carry out upstream processing and downstream analysis across different microbiome study designs without requiring a fixed end-to-end workflow template.

1 Introduction

Advances in sequencing have expanded opportunities to investigate microbial communities in human, plant, and environmental systems. Analyses

of 16S rRNA gene amplicons and shotgun metagenomes can characterize community composition, reveal functional potential, and recover microbial genomes [1]. Turning these data into biological insights, however, requires researchers to connect sequence processing with taxonomic and functional annotation and downstream statistical analysis. Selecting suitable methods and reference databases, configuring software, writing analysis code, and interpreting the results require both computational skills and biological knowledge. These demands create substantial barriers for researchers without specialist bioinformatics training, limiting their ability to conduct and extend microbiome analyses independently [2, 3].

Existing analysis platforms and workflow tools address parts of this burden. Workflow managers such as Snakemake and Nextflow organize computational dependencies and support reproducible, scalable execution [4, 5]. Once an analysis has been specified, these systems can coordinate its computational steps. Researchers must still determine which analyses address their question, configure the required resources, and interpret the outputs. Adding a new comparison or following up an intermediate finding can also require changes to the workflow or additional code. Reducing this burden therefore requires support for planning and revising analyses as well as executing them.

Large language model agents offer a way to connect these decisions with computational execution through natural-language interaction. AutoBA combined analysis planning, code generation, execution, and automated repair for multi-omics tasks, while Biomni integrated tools and resources for a broad range of biomedical analyses [6, 7]. BioMedAgent, BioMaster, and M3A have further explored automated bioinformatics and human supervision of multi-step analyses [8–10]. For microbiome researchers, the practical need is to carry a study objective through an entire analysis, from processing sequencing reads to comparing communities or examining the functions of individual organisms. Meeting this need requires the agent to select appropriate methods for the input data and connect their outputs to subsequent analyses without requiring users to write commands or assemble the workflow themselves.

MetaClaw is an AI agent designed specifically for microbiome analysis. It connects predefined upstream workflows on FlowHub with customizable downstream analyses through a YAML pipeline registry [11]. This architecture brings upstream processing and downstream interpretation into one agent-guided analysis, while retaining registered workflows and an external platform for upstream metagenomic processing. During execution, MetaClaw manages long-running tasks in the background and monitors their progress through repeated status checks within the active conversation. For researchers who need to combine methods beyond registered workflows or process sequencing data on their own infrastructure, flexible composition and local execution of both upstream and downstream analyses remain im-

portant requirements. Flexibility during execution also depends on whether users can revise pending analyses while a computation is running. Keeping the conversation occupied with status checks limits this interaction, motivating an execution mechanism that allows background computation and user feedback to proceed independently.

Here we present MicroBioAgent, an AI-agent framework that translates research objectives into locally executable microbiome analyses. It combines individual analytical skills for 16S amplicon analysis, reference-based metagenomics, and assembly-based metagenomics with downstream statistical analysis and functional annotation, without requiring a predefined end-to-end workflow template. An analysis plan is established before execution and revised in response to intermediate findings or user feedback. Configurable software environments and reference resources support local upstream and downstream execution without uploading raw sequencing reads to an external analysis platform. During long-running computations, the task continues in the background while the conversation remains available for user feedback. The agent can update pending analyses and automatically continue execution after the background task finishes. We evaluated its analytical capabilities through reanalyses of three published datasets comprising 230 sequencing libraries, examining completion of the requested analyses and recovery of reported biological patterns despite differences in methods and databases. To assess robustness, we evaluated task completion and output consistency across three language models and three levels of prompt detail (three replicates per combination; 27 independent sessions). In addition, ablation experiments comparing the system with and without the runtime patch assessed end-to-end task completion, method adherence, and token consumption, as well as whether users could revise pending analyses while a background task was running and whether the agent automatically resumed analysis after the task finished. MicroBioAgent aims to make microbiome analysis more accessible to researchers without specialist bioinformatics training by reducing manual coding, tool integration, and supervision of long-running computation while preserving flexibility in method selection and user control.

2 Results

2.1 Composing study-specific analyses across microbiome workflows

MicroBioAgent is an AI-agent framework that integrates flexible composition of upstream and downstream microbiome analyses, local execution, and asynchronous human-in-the-loop interaction (figure 1). Given a research question expressed in natural language, it develops an analysis plan and selects microbiome skills to carry it out. These skills provide method-specific

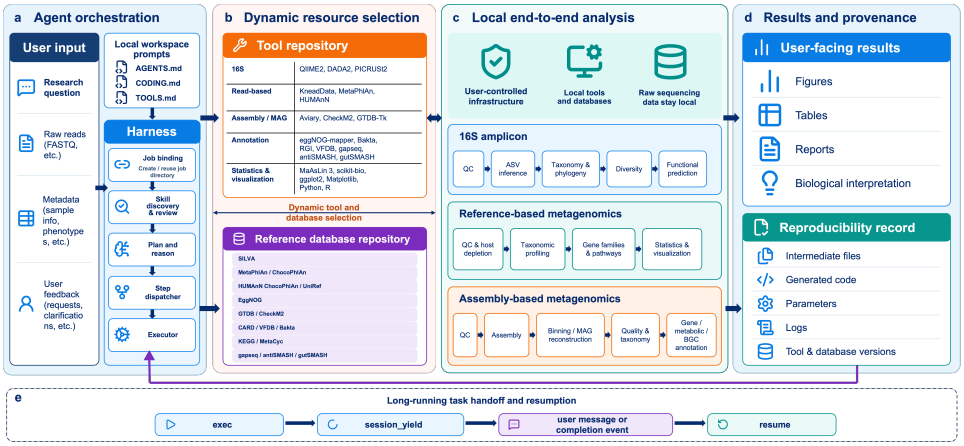


Figure 1: MicroBioAgent architecture and operating model. A user request is bound to an analysis job, from which the agent plans, selects skills, and resolves local runtimes and reference resources for 16S amplicon, reference-based shotgun, and assembly-based metagenomic analyses. Intermediate artifacts, commands, parameters, and logs remain associated with the job. During long-running computation, a user message can update pending work, whereas command completion triggers output verification and continuation in the same session.

guidance that the agent draws on when choosing analytical approaches, preparing commands, and interpreting outputs. As each step completes, the agent examines its results and decides whether the pending plan needs adjustment.

MicroBioAgent executes upstream processing and downstream analyses in a user-controlled local computational environment, without requiring raw sequencing data to be uploaded to an external analysis platform. To support these analyses, the system manages analytical tools and reference databases through separate configuration files. The tool configuration specifies available software and its associated execution environments, while the database configuration records reference resources and their local storage paths. Users can modify these configurations to add or replace tools and reference databases according to their research needs.

For upstream analysis, MicroBioAgent supports 16S amplicon processing, reference-based metagenomic profiling, and assembly-based metagenomic reconstruction (table 1). Its amplicon capabilities include sequence quality assessment, denoising, amplicon sequence variant inference, taxonomic classification, and phylogenetic reconstruction. For shotgun data, it supports read quality control and host-read removal, taxonomic and functional profiling, and gene-catalogue construction. Assembly-based capabilities include metagenomic assembly and binning, MAG quality assessment, dereplication,

taxonomic assignment, and abundance estimation.

MicroBioAgent supports a range of downstream analyses, such as comparisons of microbial communities and characterization of the functions of individual organisms. These include comparisons of diversity and taxonomic or functional abundances between groups, covariate association testing, functional enrichment, association network analysis, and supervised classification. For genes and recovered genomes, available analyses include functional and pathway annotation, antimicrobial-resistance and virulence-gene annotation, metabolic reconstruction, and identification of primary metabolic and secondary-metabolite biosynthetic gene clusters. For visualization, MicroBioAgent supports ordination plots of community differences and heatmaps of taxonomic abundances or functional profiles, with customization to highlight the comparisons of interest.

During long-running computations, such as MetaPhlAn or HUMAnN profiling, MicroBioAgent keeps the conversation available for user interaction while the analysis runs in the background. Users can provide additional instructions or revise the pending analysis plan without waiting for the running task to finish. This asynchronous human-in-the-loop interaction allows the agent to incorporate feedback during execution and automatically resume the analysis under the updated plan after checking the completed task's outcome. MicroBioAgent thus supports end-to-end microbiome analysis through flexible workflow composition and local execution, while allowing users to guide ongoing analyses.

2.2 Study-specific analyses across plant, human, and environmental microbiomes

We evaluated MicroBioAgent on three published datasets comprising 230 sequencing libraries from plant, human, and environmental microbiomes (Supplementary Table S1). These case studies assessed the system's ability to complete the requested analyses and recover biological patterns reported in the source studies. Because software versions, reference databases, and some statistical procedures differed, exact computational reproduction was not the evaluation objective. Instead, we assessed completion of the requested analyses, validity of the resulting outputs, and recovery of prespecified biological signals.

The three cases were selected to test MicroBioAgent on distinct combinations of data types and analytical tasks. The tomato rhizosphere case combined 60 16S amplicon libraries with 36 matched shotgun libraries to examine differences in microbial diversity and taxonomic composition associated with pathogen inoculation and cultivar resistance. The colorectal cancer case used 128 faecal shotgun metagenomes from 74 patients and 54 controls to compare species and functional abundances and train classifiers based on gene-family and KEGG Orthology profiles. The third case used six shotgun

Table 1: MicroBioAgent analytical capabilities, grouped by scientific task. Functional predictions and genome annotations describe inferred biological potential.

Task category	Supported analyses
Amplicon characterization	Read import and quality summaries, denoising and ASV inference, taxonomy, phylogeny, predicted functional profiles
Shotgun profiling	Read quality control and host depletion, species abundances, gene-family and pathway profiles, nonredundant gene catalogues
MAG characterization	Assembly and binning, quality assessment. Dereplication, taxonomic assignment, abundance estimation, genome annotation
Community comparisons	Alpha and beta diversity, ordination, group comparisons, differential abundance, covariate association testing
Functional interpretation	Gene and protein annotation, orthology, pathway, and ontology mapping, functional enrichment, resistance and virulence gene annotation
Metabolic characterization	Metabolic pathway and model reconstruction, primary metabolic gene clusters, secondary-metabolite biosynthetic gene clusters
Study-specific analysis	Custom statistical scripts, association networks, supervised classification, figures, result tables, and analytical reports
Data and resource access	Public sequencing metadata and reference-protein retrieval, local reference resources and online annotation queries

metagenomes from soil-derived enrichment cultures grown on low-molecular-weight polyethylene or alkane substrates to generate MAGs through assembly and binning, dereplicate them, characterize their taxonomic composition, and annotate degradation-associated genes in focal genomes.

2.2.1 A tomato rhizosphere bacterial-wilt resistance study

We applied MicroBioAgent to a published study of tomato rhizosphere microbiomes associated with resistance to bacterial wilt [12]. This case evaluated its ability to connect amplicon and shotgun analyses to examine resistance-associated community changes and identify the species markers reported in that study. The input comprised 60 16S V3–V4 amplicon libraries and 36 matched shotgun libraries from two susceptible cultivars (A57 and HG70), a resistant parent (HG64), and a resistant F1 hybrid (HF12). The amplicon data covered five sampling conditions with three biological replicates per cultivar, including before inoculation (T1), and inoculated and uninoculated plants at day 5 (T2 and T2C) and day 10 (T3 and T3C). The matched shotgun libraries covered T1, T2C, and T2. Starting from sequencing reads, MicroBioAgent performed quality control, DADA2 denoising and taxonomic classification against the SILVA database for the amplicon data, and quality control, host-read removal and MetaPhlAn3 taxonomic profiling for the shotgun data. The resulting profiles supported community-level comparisons using 16S data and species-level marker analysis using shotgun data.

For the 16S amplicon data, MicroBioAgent examined how pathogen inoculation and cultivar resistance were reflected in microbial diversity and community composition. Following *Ralstonia* inoculation, Shannon diversity declined in susceptible cultivars and remained higher in resistant cultivars at both day 5 ($P = 0.002$) and day 10 ($P = 0.004$, figure 2a). Community composition also differed between inoculated and uninoculated samples, which separated along the first weighted-UniFrac PCoA axis (56.3% of variance). Resistant and susceptible samples showed a less pronounced separation along the second axis (12.2%), with overlap between the two groups (figure 2b). Genus-level profiles linked these community changes to an expansion of *Ralstonia*, most pronounced in the susceptible cultivars A57 and HG70 (figure 2c). These patterns were consistent with the inoculation-associated community changes and higher post-inoculation diversity in resistant cultivars reported in the source study.

To examine the taxa underlying these resistance-associated differences, MicroBioAgent compared each resistant cultivar with the pooled susceptible cultivars using MaAsLin 3. In the 16S profiles, the number of genera enriched in both HG64 and HF12 ranged from 6 to 15 across sampling conditions, with different genera contributing to these shared enrichments (Supplementary Figure S1). The matched shotgun data provided species-level comparisons of the markers highlighted in the source study. MetaPhlAn3

profiles showed enrichment of the *Pseudomonas putida* group in both the resistant parent HG64 ($P = 0.013$) and the resistant hybrid HF12 ($P = 0.009$) at T1. The other reported marker, *Sphingopyxis unclassified*, was not identified in the MetaPhlAn3 profiles. Following investigator review, MicroBioAgent repeated the shotgun profiling using MetaPhlAn2, the version used in the original study, and identified both *Pseudomonas putida* and *Sphingopyxis unclassified* as enriched in resistant cultivars, recovering both reported taxonomic associations.

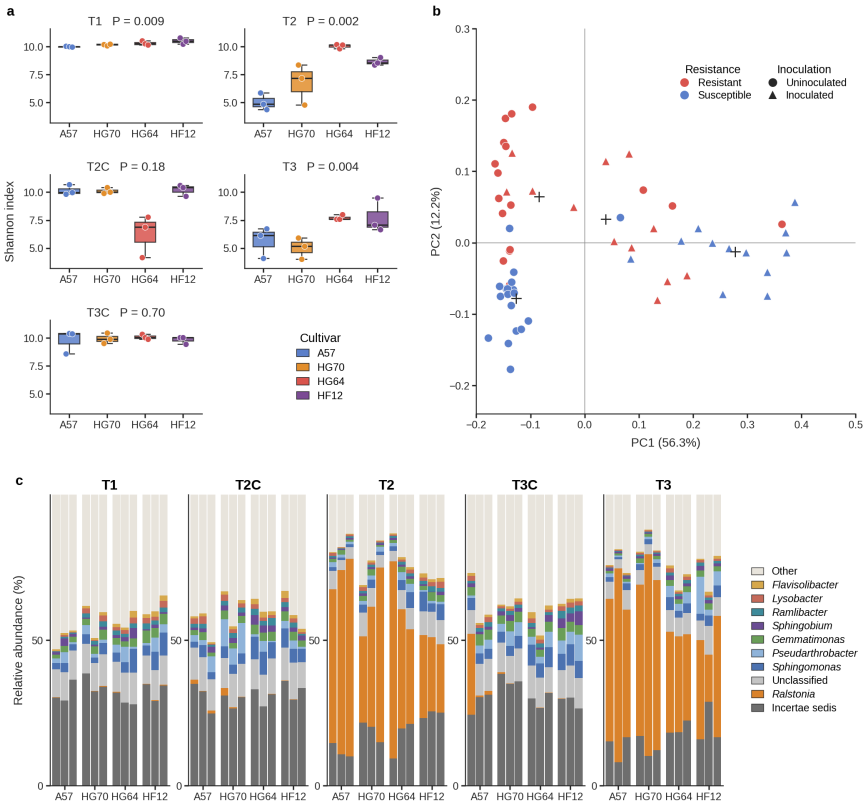


Figure 2: Tomato rhizosphere diversity and community composition. (a) Shannon diversity across cultivars, sampling time points, and inoculation conditions. (b) Weighted-UniFrac PCoA, coloured by resistance phenotype and shaped by inoculation treatment. (c) Genus-level composition across individual samples. Each cultivar-time-point combination contained three biological replicates.

2.2.2 A colorectal cancer faecal metagenomic biomarker study

The colorectal cancer (CRC) case evaluated whether MicroBioAgent could connect taxonomic and functional profiling to disease-associated species anal-

ysis, co-occurrence networks and predictive modelling. The dataset comprised 128 faecal metagenomes from the Hong Kong C1 cohort reported by Yu *et al.*, including 74 patients with CRC and 54 controls [13]. MicroBioAgent performed species-level profiling with MetaPhlAn3 and quantified UniRef90 gene-family and KEGG Orthology (KO) abundances with HUMAnN3 for downstream association analysis and classification (figure 3). Reported species associations and co-occurrence patterns provided biological reference points, while cross-validation assessed the predictive performance of the functional features.

PCoA based on Bray–Curtis dissimilarities calculated from MetaPhlAn3 species-level relative abundance profiles showed no clear separation among controls, early-stage CRC cases ($n = 36$) and late-stage CRC cases ($n = 38$). PERMANOVA nevertheless detected differences in community composition associated with CRC status ($F = 2.32$, $P = 0.002$) and disease stage ($F = 1.65$, $P = 0.009$). Separate tests for sex, age and BMI were not significant.

Comparisons of species abundances between cases and controls identified eight CRC-enriched and two control-enriched taxa after Benjamini–Hochberg correction ($q < 0.05$). Four CRC-enriched species, *Parvimonas micra*, *Peptostreptococcus stomatis*, *Fusobacterium nucleatum* and *Solobacterium moorei*, showed the same direction of association as in the original study. Enrichment of *Eubacterium ventriosum* in controls also agreed with the reported association. To examine whether these disease-associated species also varied together across samples, Spearman co-occurrence analysis was performed on the ten differentially abundant species. The network contained 16 positive correlations ($|\rho| \geq 0.4$, BH-adjusted $q < 0.05$), with *Parvimonas micra* connected to six other species. Its positive correlations with *Fusobacterium nucleatum* ($\rho = 0.49$) and *Solobacterium moorei* ($\rho = 0.53$) were consistent with the co-occurrence relationships emphasized in the original study [13].

The analysis then extended from identifying disease-associated taxa to testing whether functional differences could distinguish CRC cases from controls. MicroBioAgent selected differentially abundant features and trained separate gene-family and KO classifiers to compare the predictive performance of these two functional representations. Within each training fold, mRMR selection yielded 15 gene-family features or 15 KO features for logistic-regression modelling. Performance was evaluated by nested fivefold cross-validation within the C1 cohort. The gene-family classifier achieved a pooled out-of-fold AUROC of 0.902, compared with 0.632 for the KO classifier, indicating stronger discrimination using gene-family features. At thresholds selected using Youden's index, the gene-family classifier achieved a sensitivity of 0.82 and specificity of 0.89, compared with 0.41 and 0.87 for the KO classifier.

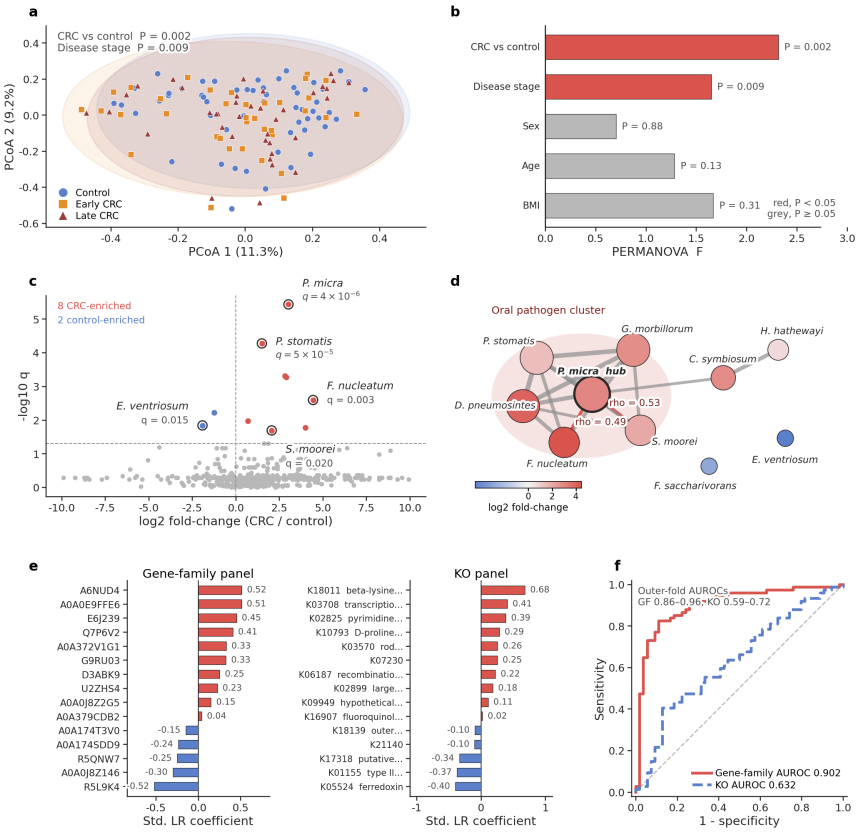


Figure 3: Reference-based analysis of CRC faecal metagenomes. Species-level beta diversity, disease-associated taxa, co-occurrence of the differentially abundant species, functional features, and nested cross-validation of gene-family and KO classifiers are shown.

2.2.3 A polyethylene-enrichment genome-resolved metagenomic study

We next evaluated whether MicroBioAgent could reconstruct genomes from enrichment metagenomes and use them to identify microorganisms and annotate genes associated with alkane and oxidized-derivative metabolism. The dataset comprised six Oxford Nanopore long-read metagenomes from two soil inocula (S1 and S2), each enriched on low-molecular-weight polyethylene (LMWPE), triacontane (C30) or tetracontane (C40) [14]. Assembly and binning generated 372 bins, of which 153 passed completeness and contamination filtering (100 high-quality and 53 medium-quality MAGs). Dereplication at 95% average nucleotide identity retained 75 representative genomes (50 high quality and 25 medium quality, figure 4a). These genomes were classified using GTDB-Tk with GTDB release R232, and their abundances across the enrichment cultures were estimated by read mapping with CoverM to identify the predominant microorganisms in each enrichment culture.

MAG abundance profiles showed different dominant taxa across the enrichment cultures. *Acinetobacter guillouiae* dominated S1 LMWPE and S1 C30, accounting for 64% and 67% of the reported abundance, respectively. S1 C40 was instead dominated by *Aquipseudomonas*, which accounted for 66% at the genus level (figure 4b,c). The S2 cultures showed a different composition. *Aquipseudomonas* sp003050835 was most abundant in S2 LMWPE (42%), while *Mesorhizobium_M* sp900116595 and *Aquipseudomonas moorei* were the most abundant species in S2 C30 (16% each). In S2 C40, *Mesorhizobium_M* sp900116595 and *Aquipseudomonas fluorescens_BJ* predominated at 27% and 25%, respectively. Taxonomic annotation identified the ten taxonomic groups of particular interest in the original study (Supplementary Data 1). The groups reported as *Pseudomonas* in the original study were classified as *Aquipseudomonas* in the present analysis due to differences in the reference taxonomy.

To examine the metabolic potential of the ten taxonomic groups emphasized in the original study, functional annotation was performed on one representative MAG from each group. The annotations identified genes associated with the three metabolic processes examined in the original study (figure 4d). Genes associated with alkane terminal oxidation were annotated in eight MAGs, while those associated with 2-ketone or oxidized-derivative metabolism were detected in all ten. Nine MAGs contained genes assigned to the β-oxidation module M00087. Eight MAGs contained annotations spanning all three categories, indicating that the corresponding genetic potential was distributed across multiple taxa of interest.

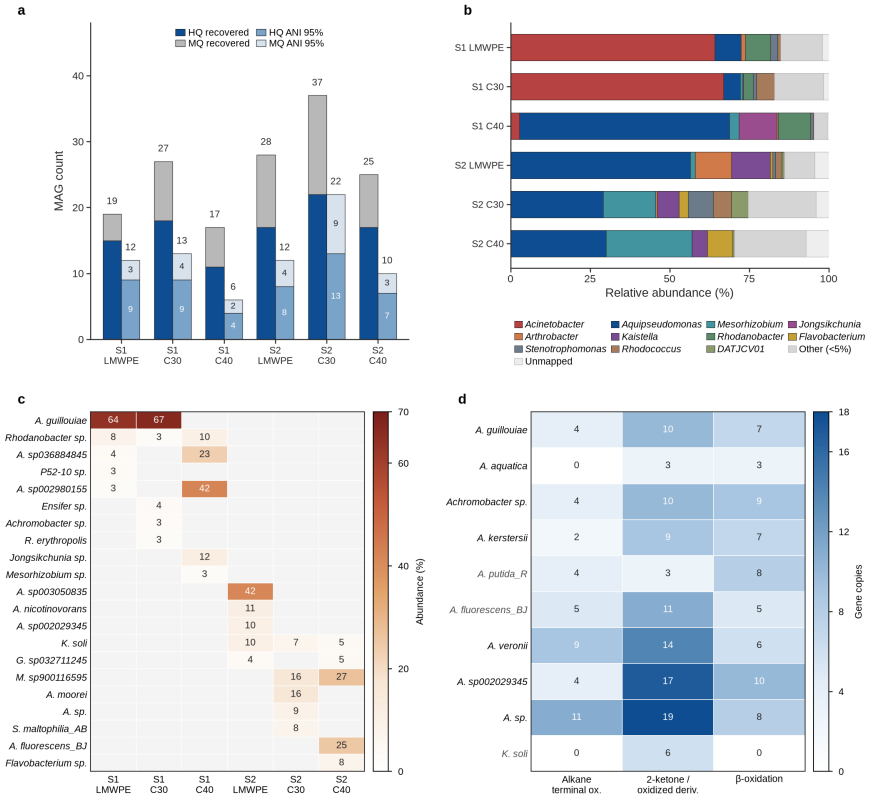


Figure 4: Assembly-based analysis of enrichment metagenomes. (a) Numbers of high- and medium-quality MAGs, grouped by completeness and contamination thresholds defined in Methods, before and after 95% ANI dereplication. (b) Genus-level MAG composition. (c) Species among the five most abundant MAGs in at least one sample. (d) Numbers of genes annotated using eggNOG-mapper in one representative MAG per target taxon for alkane terminal oxidation, 2-ketone or oxidized-derivative metabolism, and β-oxidation (M00087). Counts represent annotated genes rather than distinct functional types or transcript abundance.

2.3 Robustness across models and prompt granularity

To assess whether MicroBioAgent maintains reliable execution and consistent analytical outputs across language models and levels of user guidance, we crossed three models—`deepseek-v4-pro`, `deepseek-v4-flash` and `glm5.3-flash`—with detailed, medium and simple prompts on the preprocessed Case 1 dataset ($3 \times 3 \times 3 = 27$ independent sessions). Across six core analyses (16S alpha diversity, 16S beta diversity, 16S genus-level taxonomic profiling, 16S differential abundance, shotgun species-level taxonomic profiling and shotgun differential abundance), MicroBioAgent completed 160 of 162 evaluations (98.8%; Supplementary Figure S2a). `deepseek-v4-pro` and `deepseek-v4-flash` completed all six analyses in three of three sessions under every prompt level, and `glm5.3-flash` completed all 16S analyses (9/9 sessions) and 8 of 9 shotgun sessions, with one session under the detailed prompt not analysing the shotgun libraries.

Quantitative outputs for community diversity and differential abundance were consistent across both models and prompt levels (Supplementary Figures S2b and S3). Per-sample 16S Shannon indices (median 9.59–9.83) and pairwise weighted UniFrac distances (median 0.301–0.303) did not differ between any pair of models under any prompt level or across prompt levels within any model (all Holm-adjusted $P = 1$, two-sided Mann–Whitney U test). For differential abundance, because none of the prompts specified a statistical test, individual sessions chose among DESeq2, MaAsLin 3, Wilcoxon rank-sum tests, Mann–Whitney U tests and LEfSe (Supplementary Table S4, panels a and b). Because different tests define effect sizes on different scales, we compared $\log_2$ fold-change distributions across sessions that used the same test. Across models within each prompt level, DESeq2 was used by all three models for 16S analyses under the detailed and medium prompts and for shotgun analyses under the detailed prompt, and its $\log_2$ fold-change distributions did not differ between any pair of models (all Holm-adjusted $P \geq 0.059$; under the medium prompt in shotgun analyses, no single test was used by all three models). Between the detailed and medium prompts within each model, $\log_2$ fold-change distributions also did not differ for any model in either 16S analyses using DESeq2 (`deepseek-v4-pro`, $P = 0.54$; `deepseek-v4-flash`, $P = 0.50$; `glm5.3-flash`, $P = 0.13$) or shotgun analyses using the test shared within each model (`deepseek-v4-pro`, Wilcoxon rank-sum test, $P = 0.58$; `deepseek-v4-flash`, Mann–Whitney U test, $P = 0.24$; `glm5.3-flash`, DESeq2, $P = 0.064$).

For taxonomic profiling, relative abundances were more sensitive to model and prompt choices (Supplementary Figures S2b and S3). Across models, relative-abundance distributions did not differ in 16S analyses under the simple prompt or in shotgun analyses under the detailed prompt (all Holm-adjusted $P = 1$), but differed in the remaining prompt settings (Holm-adjusted $P \leq 0.05$). Across prompt levels within each model, distributions

were consistent across all three prompts for `glm5.3-flash` in 16S analyses and for both `deepseek-v4-pro` and `deepseek-v4-flash` in shotgun analyses (all Holm-adjusted $P = 1$), whereas differences occurred in the remaining comparisons (Holm-adjusted $P \leq 0.025$). Inspecting the session logs showed three causes of these differences: (1) some sessions applied a V4 rather than a V3–V4 classifier during 16S taxonomic classification; (2) an occasional script misaligned feature identifiers when merging sample profiles into a taxonomic abundance table; and (3) without prompt constraints on MetaPhlAn parameters, sessions differed in whether `-unknown_estimation` was enabled, changing whether species relative abundances were normalized to total reads or only classified reads. Overall, MicroBioAgent maintained high task completion and consistent diversity and differential-abundance outputs across models and prompt levels, with remaining differences largely reflecting choices of analytical tools and software parameters.

2.4 Evaluating asynchronous execution and guidance

2.4.1 End-to-end execution and token consumption

To evaluate how background-execution coordination affects end-to-end analysis, we compared the complete system (`full`) against an ablated configuration without the runtime patch (`no-patch`; table 2 and section 4.1.3). Both conditions used `deepseek-v4-pro`, the detailed Case 1 prompt, and the preprocessed Case 1 dataset across three independent sessions, with no user messages after initial plan confirmation.

Across the three sessions, the complete system produced all 27 requested outputs (9 of 9 per session) and followed all 24 prespecified methodological requirements (8 of 8 per session). The `no-patch` configuration followed all 24 methodological requirements and produced 26 of 27 requested outputs, with one session omitting archived analysis scripts. Removing asynchronous execution coordination therefore did not prevent the model from completing the prespecified downstream workflow when the prompt explicitly named the required methods, but it substantially altered how long-running commands were managed.

In the complete system, once a long-running command was moved to the background, the agent paused its conversation turn (via the `sessions_yield` tool, mean 11.7 calls per session, range 5–17) to await automatic notification upon command completion, and rarely queried whether the background process had finished (via the `process` tool, mean 7.3 calls, range 0–21, with two sessions making at most one call). In `no-patch`, by contrast, the agent never paused to await completion notifications (0 `sessions_yield` calls across all three sessions) and instead repeatedly queried the status of running commands with `process` (mean 155.0 calls per session, range 102–227). Because each status check required a new model call that re-read the grow-

ing conversation history, total model calls increased from a mean of 133.3 (range 124–150) in `full` to 286.3 (range 211–403) in `no-patch`, and mean total token consumption rose by 141.5% (2.4-fold), from 14.53×10^6 (range $12.92–15.84 \times 10^6$) to 35.08×10^6 (range $23.77–49.32 \times 10^6$) tokens per session (table 2).

Table 2: End-to-end execution and token consumption with and without the runtime patch.

Condition	Session	Outputs (of 9)	Methods (of 8)	Model calls	`exec` calls	`process` polls	`yield` calls	Total tokens ($\times 10^6$)
`full`	Replicate 1	9/9	8/8	150	91	1	17	12.92
`full`	Replicate 2	9/9	8/8	124	74	21	5	14.81
`full`	Replicate 3	9/9	8/8	126	82	0	13	15.84
`full`	Mean / sum	27/27	24/24	133.3	82.3	7.3	11.7	14.53
`no-patch`	Replicate 1	8/9	8/8	211	77	102	0	23.77
`no-patch`	Replicate 2	9/9	8/8	403	142	227	0	49.32
`no-patch`	Replicate 3	9/9	8/8	245	99	136	0	32.15
`no-patch`	Mean / sum	26/27	24/24	286.3	106.0	155.0	0.0	35.08

Notes: `full` denotes the complete system with the runtime patch, and `no-patch` denotes the ablated configuration without the patch. All sessions used `deepseek-v4-pro`, the detailed Case 1 prompt, and the preprocessed Case 1 dataset. Outputs and methods in the summary rows are pooled counts across the three sessions; call counts and token values in the summary rows are per-session means. Total tokens equal input plus output tokens (including reasoning tokens).

2.4.2 Asynchronous human guidance during active computation

We next tested whether the user could revise the pending analysis plan while a minutes-scale Case 1 QIIME 2 taxonomy-classification command was running in the background. Once the classification process was confirmed active, the user sent an instruction to keep the running classification command and update the pending plan to use DESeq2 for the subsequent differential-abundance analysis and generate a heatmap of significant genera.

In the complete system (`full`), after starting the classification task, the agent paused its conversation turn via `sessions_yield` and updated the pending plan in response to the user's instruction while the task was still running (3/3 sessions; table 3). Once the classification task finished, the session was automatically reactivated to continue the analysis without additional user input or repeated `process` checks (3/3 sessions). In `no-patch`, by contrast, because the agent did not pause its turn via `sessions_yield` and instead kept the conversation occupied with repeated `process` calls to check whether the task had completed, neither mid-run plan revision nor automatic session reactivation upon command completion was achieved in

any of the three sessions (0/3).

Table 3: Ablation of asynchronous human guidance and automatic session reactivation during active computation. Each condition was evaluated in three independent sessions.

Condition	Plan updated while command running	Automatic reactivation upon command completion
`full`	3/3	3/3
`no-patch`	0/3	0/3

Notes: The two conditions compare the complete system (`full`) with the ablated configuration without the runtime patch (`no-patch`; section 4.1.3). Plan updated while command running requires the agent to incorporate the user's mid-run instruction into the pending analysis plan while the classification task is still running in the background. Automatic reactivation upon command completion requires completion of the background classification command to automatically reactivate the waiting session so that the agent continues with the revised plan without additional user messages or repeated `process` calls to check whether the command has completed. Results are reported as successful sessions out of three.

3 Discussion

MicroBioAgent combines modular domain skills, local execution environments, and asynchronous human-in-the-loop interaction to translate biological research questions into executable microbiome analyses. This framework is supported by three complementary lines of evidence: (1) three case studies across plant, human, and environmental microbiomes demonstrating flexible composition across amplicon, shotgun metagenomic, and genome-resolved workflows; (2) 27 independent sessions across three language models and three prompt granularities establishing execution robustness and quantitative output consistency; and (3) ablation experiments demonstrating that the runtime patch substantially reduces token overhead while enabling mid-run plan revision and automatic continuation. By coordinating established bioinformatics tools within an inspectable and interactive workflow, MicroBioAgent allows researchers to transparently steer complex analyses without manual coding or fixed pipeline constraints.

The three case studies demonstrate how MicroBioAgent adapts to distinct research questions, spanning cultivar-associated marker discovery in paired amplicon and shotgun libraries, disease-predictive modeling from functional profiles, and genome-resolved metabolic reconstruction. At the same time, these reanalyses show that biological findings remain sensitive to statistical thresholds, tool configurations, and reference database versions. In the tomato rhizosphere study, initial profiling with MetaPhlAn3 detected only one of the two reported species markers, whereas repeating the analysis with MetaPhlAn2, the version used in the original study, recovered

both. By preserving commands, parameters, and intermediate tables within the session context, MicroBioAgent makes these analytical choices explicit, allowing researchers to inspect the results and adjust the workflow when needed.

MicroBioAgent occupies a distinct position within the growing landscape of AI agents and workflow tools for biomedical research (Supplementary Table S3). General-purpose biomedical agents, such as Biomni, combine tool retrieval with code-based execution across diverse domains [7], but lack domain-specific skill guidance and configured reference databases required for end-to-end microbiome analyses beginning from raw sequencing reads. Within the microbiome domain, MetaClaw spans upstream and downstream analyses by coupling predefined FlowHub workflows with customizable downstream skills [11]. In contrast, MicroBioAgent unifies both upstream processing and downstream analysis into modular skills that run entirely on the user's local infrastructure. This on-premises architecture keeps raw sequencing data secure, eliminates dependencies on external computing platforms, and allows method choices to be inspected and adjusted at any stage. Such an interactive approach complements workflow managers such as Nextflow or Snakemake rather than replacing them. While traditional workflow engines excel at large-scale, deterministic execution of static pipelines, MicroBioAgent is tailored for exploratory and iterative analyses where plans could be guided and refined in response to intermediate results.

The cross-model and cross-prompt benchmark illustrates the effectiveness of MicroBioAgent in automating complex microbiome workflows. By generating custom scripts, switching software environments, and coordinating input–output handoffs across tools, the system achieved high overall task completion (98.8%) and consistent diversity and differential-abundance distributions across models and prompt granularities on the preprocessed Case 1 dataset. Nevertheless, occasional inaccuracies in model reasoning also occurred, including misidentifying the V3–V4 amplicon region as V4, altering MetaPhlAn normalization flags (`-unknown_estimation`), and merging feature tables by row position rather than matching identifiers. As foundation model capabilities continue to advance, such procedural inconsistencies are expected to diminish. More fundamentally, these findings highlight the practical value of MicroBioAgent's transparent architecture. By automating end-to-end computational execution while keeping intermediate commands, parameters, and outputs fully inspectable, the system enables researchers to readily verify analytical decisions and refine workflows without constructing them manually.

The ablation experiments evaluate the operational contribution of the runtime patch during long-running microbiome computations (tables 2 and 3). In conventional agent setups without the patch, monitoring an active process occupies the conversational turn with repeated status checks, which increased mean model calls from 133.3 to 286.3 and inflated total token con-

sumption by 2.4-fold (from 14.53×10^6 to 35.08×10^6 tokens per session) due to continuous re-reading of the expanding conversation history. More importantly, keeping the session busy with status polling blocks user interaction, preventing researchers from providing timely guidance while computations are underway. By pausing the conversation turn and relying on completion events to reactivate the session, the runtime patch eliminates status-polling overhead while enabling reliable mid-run plan revision and automatic continuation. This capability is especially valuable for multi-step microbiome analyses, where heavy operations such as sequence denoising, taxonomic classification, and metagenomic assembly often run for hours while researchers review intermediate findings and refine downstream hypotheses.

While the case studies and benchmarks show that MicroBioAgent can support interactive microbiome analyses, several practical limits remain. First, our robustness and ablation experiments used the preprocessed Case 1 dataset, and mid-run guidance was tested on a minutes-scale 16S taxonomic classification step across three replicates per condition. Testing the system on multi-hour assembly pipelines, broader command failures, and more detailed plan changes remains an important next step. Second, automated execution checks do not guarantee that biological findings are correctly interpreted, so researchers still need to inspect intermediate tables and confirm overall analysis plans. Third, running analyses locally protects sensitive sequencing data, but users must still install and maintain the required tools, Conda environments, and reference databases on their own machines. Future work should include direct benchmark comparisons with a general coding agent to isolate the practical contribution of preconfigured domain skills and runtime setups. In addition, formal user studies will be valuable to assess whether this conversational framework measurably reduces operational time and cognitive burden for life scientists. Finally, while this study focused on amplicon and metagenomic data, the core architecture of combining modular skills, local environments, and background execution can be extended to other areas, such as transcriptomics or metabolomics, by adding new skill guides and software environments. Overall, MicroBioAgent is not intended to replace human expertise, but to provide a transparent assistant that handles repetitive bioinformatics execution while leaving researchers in full control of analytical decisions.

4 Methods

4.1 Details on implementing MicroBioAgent

MicroBioAgent was implemented on OpenClaw to plan and execute microbiome analyses from natural-language requests (figure 1). Users communicate with the system through a web interface, a command-line client, or supported messaging applications such as WhatsApp, Slack and Telegram.

These interfaces send user requests to the OpenClaw gateway, which maintains the conversation and passes the messages to the language model. When the model requests a tool operation, the gateway invokes the corresponding tool. For computational analyses, bioinformatics commands and statistical scripts run on the user's workstation or server, using software installed directly on the host or within a docker container. Execution status and captured output return to the conversation, allowing the model to determine subsequent actions.

The behaviour of MicroBioAgent is guided by Markdown instruction files in its agent workspace. `AGENTS.md` governs task planning and progress tracking, while `TOOLS.md` describes access to software and reference resources. `CODING.md` guides code execution and output inspection. Method-specific skills provide analytical guidance, and separate tool and database configurations identify the resources used to execute each step. Supporting user interaction during long-running steps also requires coordination between command execution and the conversation. We therefore introduced a patch to the OpenClaw runtime that modifies tool-execution guidance and command-completion handling. The patch and workspace instructions allow computational tasks to continue in the background while the agent responds to user feedback. When a background task finishes, the agent is automatically reactivated to inspect its outputs and continue the remaining analysis.

4.1.1 Analytical skills and configured execution

Analytical methods are organized into skills covering 16S amplicon processing, reference-based metagenomic profiling, assembly-based genome reconstruction, and downstream statistical analysis, annotation and visualization (table 1). Each skill contains a `SKILL.md` document describing its purpose, required inputs, software dependencies, reference resources, analytical procedures and expected outputs. Some skills also include reference scripts that can be adapted to the study data. These skills are registered through the plugin manifest, which defines their names, locations and metadata, and are made available through OpenClaw's skill mechanism. For each analytical step, MicroBioAgent selects a relevant skill and reads its `SKILL.md` through the file-reading tool, bringing the method-specific guidance into the model's context. This stepwise reading is guided by `AGENTS.md`, which specifies that the relevant skill should be consulted before execution.

Software environments and reference database locations are recorded in two configuration files. The `runtimes.yaml` file identifies the Conda environment containing each tool or library, whereas `references.yaml` specifies the locations of the reference databases. The reference configuration is generated from user-supplied settings when the OpenClaw gateway starts, with database paths adjusted for its running environment. Following the resource-access guidance in `TOOLS.md`, the agent would read these configurations to

select the required software environment and locate the reference databases. Together, modular skills and separate resource configurations allow users to extend the available analytical methods. A new method can be introduced through an additional skill that uses existing software environments and reference databases or specifies additional resources in the corresponding configuration files.

4.1.2 Analysis planning and execution

MicroBioAgent uses `mgx_create_job` to create or reuse an analysis directory for each conversation. The tool records the association between the conversation identifier and the directory on disk, allowing successive steps to share the same locations for inputs, outputs, scripts and logs. With the analysis directory established, MicroBioAgent develops a plan based on the research objective, input data and requested outputs. The plan defines the analytical steps, their order and completion criteria, and is maintained within the conversation. Method-specific skills provide methodological guidance for the planned analyses, while `CODING.md` guides script development and command execution through `exec`. The execution guidance standardizes agent script authoring rules to ensure operational errors are promptly captured and handled, while enabling the agent to execute independent tasks in parallel and balance concurrency and thread counts against available CPU and memory resources. It also specifies how execution status, logs and generated files are checked after a command launched through `exec` finishes. These checks determine whether execution succeeded and whether the expected output files were produced. If either condition is not met, the agent reviews the logs and outputs to identify the problem and determine which parts of the analysis need to be rerun.

4.1.3 Human-in-the-loop interaction and automatic continuation

MicroBioAgent allows users to interact with the agent while computational tasks run asynchronously in the background. Guided by `CODING.md` and tool instructions, the agent executes commands via `exec` with a 30-second initial wait period and no overall execution time limit. Commands exceeding this threshold continue running in the background, prompting the agent to invoke `sessions_yield` to yield its turn. MicroBioAgent handles these background processes and logs their outputs without requiring the agent to poll for status updates. This decoupling keeps the conversation responsive, enabling users to discuss or revise pending analyses without interrupting running computations. Both user messages and command-completion notifications can trigger the agent within the same thread, supporting seamless continuous interaction and automatic task continuation upon completion.

To enable automatic continuation after background execution, we de-

veloped a patch for OpenClaw (v2026.7.2) that extends its event handling and conversation activation mechanisms. Each background command is associated with its originating conversation ID, ensuring completion events are dispatched to the corresponding conversation. For an idle conversation, the patch immediately triggers a new agent loop upon command completion without waiting for scheduled polling or user input. If an event arrives during an active agent run, it is queued and schedules a subsequent activation. This queuing mechanism prevents command results from being dropped when the agent is busy, ensuring they are reliably processed in the next turn. The injected event message contains the command result along with guidance instructions to prompt further analysis. Together with updated tool instructions that guide the agent to yield its turn, these changes allow background computation to run alongside active conversation and support automatic continuation upon task completion.

4.2 Case study design

To assess MicroBioAgent across diverse microbiome workflows, we evaluated the system on three published datasets spanning plant, human, and environmental ecosystems (Supplementary Table S1), covering 16S rRNA amplicon profiling, reference-based shotgun metagenomics, and assembly-based genome reconstruction. Across all three cases, the agent was powered by `deepseek-v4-pro` and provided with sequencing reads, sample metadata, and natural-language research goals without pre-written analysis scripts. The evaluation tested whether the agent could autonomously compose end-to-end workflows and recapitulate the primary biological conclusions of the original studies using its configured software environments and reference databases. When specific metrics or domain-standard tools were explicitly requested in the prompt, the agent applied them; otherwise, all analytical procedures, software environments, and command parameters were determined autonomously using its available skills. The specific workflows and evaluation tasks for each case study are detailed below.

4.2.1 Case 1 — Tomato rhizosphere 16S and metagenomic analysis

This case evaluated MicroBioAgent's ability to analyze paired amplicon and shotgun metagenomic datasets to identify microbial signatures associated with disease resistance in tomato. Sequencing data were retrieved from BioProject PRJNA777724 [12]. The dataset comprises 60 rhizosphere 16S rRNA (V3–V4) amplicon libraries generated from four cultivars—two susceptible (A57, HG70) and two resistant (HG64, HF12). Samples were collected across five experimental conditions: baseline (pre-inoculation at day 0), as well as inoculated and uninoculated controls at days 5 and 10 ($n = 3$ biolog-

ical replicates per group). Additionally, 36 matched shotgun metagenomic libraries were profiled for the baseline (day 0) and day-5 timepoints. The evaluation comprised two analytical stages, including an initial autonomous workflow and a researcher-guided re-analysis. In the initial stage, the agent independently performed sequence processing, diversity profiling, community composition comparisons, and identification of taxa enriched in resistant cultivars relative to susceptible controls. Comparisons with the source study focused on inoculation-induced diversity shifts and biomarker identification at the genus and species levels. In the second stage, to evaluate the impact of profiler version on taxonomic recovery, researchers prompted a follow-up reanalysis using MetaPhlAn2 to align with the original study's tool rather than the agent's default MetaPhlAn3 configuration.

4.2.2 Case 2 — Colorectal cancer faecal metagenomic analysis

This case evaluated MicroBioAgent's ability to identify disease-associated microbial features and assess their predictive value for colorectal cancer (CRC). The analysis used the Hong Kong C1 cohort, available under accession PRJEB10878, comprising 128 faecal metagenomes from 74 patients with CRC and 54 controls [13]. The CRC group included 36 early-stage cases (stages I–II) and 38 late-stage cases (stages III–IV). The analysis tasks covered community diversity and composition, differential abundance of species and functional features, and associations with clinical variables. Functional analyses included both gene-family and KEGG Orthology (KO) abundance profiles. The task further required species co-occurrence network analysis to examine relationships among CRC-associated taxa, together with marker selection and classifier development to distinguish CRC cases from controls. Comparisons with the original study focused on evaluating the direction of species associations with CRC and co-occurrence patterns among disease-associated taxa, while the predictive performance of the selected markers and classifier was assessed using AUROC, sensitivity, and specificity.

4.2.3 Case 3 — Polyethylene-enrichment genome-resolved metagenomic analysis

This case evaluated MicroBioAgent's ability to recover microbial genomes from metagenomic data and characterize their functional potential. The evaluation utilized six Oxford Nanopore long-read metagenomes from project PRJEB97175 [14], derived from two soil inocula enriched on low-molecular-weight polyethylene, triacontane, or tetracontane. The analysis tasks covered metagenomic assembly, binning, quality filtration for high- and medium-quality metagenome-assembled genomes (MAGs), dereplication, and taxonomic classification, alongside cross-culture abundance profiling and gene functional annotation for MAGs assigned to ten focal taxa from the original

study. Evaluation examined whether MAGs corresponding to the ten taxa of interest in the original study were recovered and whether their abundance patterns across enrichment cultures were characterized. Functional evaluation examined whether the annotations identified genes associated with the three metabolic processes investigated in the original study, namely alkane terminal oxidation, metabolism of 2-ketones or oxidized derivatives, and β-oxidation.

4.3 Robustness and ablation experiments

To reduce computational overhead during repeated testing, robustness and ablation experiments were conducted using preprocessed data from Case 1. The 16S dataset comprised the ASV table, representative sequences, a rooted phylogenetic tree, denoising statistics, and sample metadata. For shotgun sequencing, 36 QC-filtered and host-cleared libraries were used, each subsampled to 20% of its original read depth. These preprocessed inputs were used across all experimental conditions, with each session running the analyses independently.

4.3.1 Robustness across models and prompt granularity

Robustness was evaluated across three large language models (`glm5.3-flash`, `deepseek-v4-flash`, and `deepseek-v4-pro`) and three levels of prompt detail. Each model–prompt combination was executed in three independent replicates, yielding a total of 27 experimental sessions. All three prompts requested joint 16S and shotgun metagenomic analyses of the Case 1 study, differing in how explicitly they specified the analytical steps and methods:

- **Detailed prompt:** Specified exact analytical tasks and diversity methods, covering dataset characterization, 16S alpha diversity (Shannon index after sequencing-depth normalization), 16S beta diversity (weighted and unweighted UniFrac and Bray–Curtis distances, weighted-UniFrac PCoA, PERMANOVA, and PERMDISP), 16S taxonomic and differential-abundance profiling, and shotgun species-level taxonomic and differential-abundance profiling. For both datasets, it requested comparing each resistant cultivar against the susceptible group at each sampling timepoint and identifying taxa enriched in both HG64 and HF12 within each of T1, T2C, and T2.

- **Medium prompt:** Retained the same analytical scope (dataset characterization, 16S diversity, 16S and shotgun taxonomic profiling, cultivar comparisons at each timepoint, and shared enriched taxa detection), but omitted specific diversity metrics and statistical tests, leaving method selection to the agent.

- **Simple prompt:** Stated only the high-level biological objective of identifying microbiome patterns associated with bacterial-wilt resistance across the 16S and shotgun datasets, without prescribing specific analytical steps or methods.

System configurations, analytical software, reference databases, and preprocessed inputs were held constant across all runs. Each replicate was initialized in a clean conversation session without further methodological guidance following initial plan confirmation.

To evaluate system performance across these conditions, task completion was systematically evaluated across six distinct analytical tasks: (1) 16S alpha diversity, (2) 16S beta diversity, (3) 16S genus-level taxonomic profiling, (4) 16S differential abundance, (5) shotgun species-level taxonomic profiling, and (6) shotgun differential abundance. A task was scored as completed when the session produced the corresponding quantitative output, such as a per-sample Shannon diversity table, a between-sample distance matrix, a genus- or species-level taxonomic abundance table, or a differential-abundance result table. For each model–prompt combination, the task completion rate was calculated as the proportion of completed runs out of the three independent sessions.

To evaluate output stability across models and prompt levels, quantitative results were averaged across completed replicate sessions within each model–prompt combination and compared across entries shared by the compared groups using two-sided Mann–Whitney U tests with Holm correction. For diversity analyses, per-sample Shannon indices ($n = 60$ samples) and pairwise weighted UniFrac distances ($n = 1{,}770$ unordered sample pairs) were averaged across replicates. For taxonomic profiling, abundance tables reported as percentages or raw counts were converted to per-sample relative abundances before removing unclassified or unnamed entries, with taxa absent from a replicate that included the sample entered as zero. Because microbial relative-abundance tables are highly right-skewed and contain many near-zero entries near the detection limit, minor numerical fluctuations in the low-abundance tail can disproportionately affect rank-based tests despite negligible differences in absolute abundance. To prevent these near-zero entries from dominating the comparison of taxonomic abundance tables, relative-abundance distributions were compared across shared sample and taxon entries with a session-mean relative abundance of at least 5% in every compared group, whereas all shared taxa (including low-abundance taxa) were included without an abundance threshold in the differential-abundance comparisons below.

For differential abundance, because the prompts did not prescribe a specific statistical test, sessions selected different methods (including DESeq2, MaAsLin 3, Wilcoxon rank-sum test, Mann–Whitney U test and LEfSe). Because different methods report differently defined effect sizes, $\log_2$ fold-

change distributions were compared only across sessions that used the same statistical test. Cross-model comparisons within each prompt level used sessions that ran DESeq2 (Supplementary Table S4, panel a), and within-model comparisons across prompt levels were conducted between the detailed and medium prompts, which specified the same comparisons between resistant and susceptible cultivars at each timepoint, using the statistical test shared by both prompts for each model (Supplementary Table S4, panel b). In both comparisons, each data point represented the $\log_2$ fold change of a taxon for a given resistant cultivar (HG64 or HF12) and sampling timepoint relative to the susceptible control (averaging across susceptible controls within a session when A57 and HG70 were reported separately). Values were averaged across replicate sessions without imputing missing entries as zero, and entries shared across all compared groups were retained.

4.3.2 Ablation of asynchronous execution and human guidance

To evaluate the role of the runtime patch for asynchronous execution and automatic continuation (section 4.1.3), we compared the complete system (`full`) with an ablated configuration without the runtime patch (`no-patch`) using `deepseek-v4-pro`, the detailed Case 1 prompt, and the preprocessed Case 1 dataset across three independent sessions per condition. All other system components, including microbiome domain skills, software environments, and reference databases, were held constant across both conditions.

In the end-to-end execution experiment, sessions were run with no user input after initial plan confirmation. Task completion was scored across nine required outputs (16S alpha diversity, 16S beta diversity, 16S genus-level taxonomic table, 16S differential abundance, shotgun species-level taxonomic table, shotgun differential abundance, figures, analysis scripts, and an analysis report), and method adherence was scored against the eight diversity and statistical requirements specified in the detailed prompt (sequencing-depth normalization, Shannon diversity, weighted UniFrac, unweighted UniFrac, Bray–Curtis dissimilarity, weighted-UniFrac PCoA, PERMANOVA, and PERMDISP). For each session, model call counts, tool invocations (`exec`, `process`, and `sessions_yield`), and total token consumption (input and output tokens, including reasoning tokens) were recorded from the session transcript.

In the asynchronous human-guidance experiment, we tested mid-run plan revision while the Case 1 QIIME 2 taxonomy-classification command was running in the background. Once the classification process was confirmed active, the user sent a standardized instruction to keep the running classification command and update the pending plan to use DESeq2 for the subsequent differential-abundance analysis and generate a heatmap of significant genera, with no further user input thereafter. Each session was evaluated on two outcomes: (1) whether the agent processed the user's message and updated the pending analysis plan while the classification command was

still running; and (2) whether, once the background classification command finished, the system automatically reactivated the waiting session so that the agent inspected the classification output and continued with the revised plan, without requiring additional user messages or repeated `process` calls to check whether the command had completed.

Data availability

Raw sequencing data are available from the public repositories associated with the three source studies. Sample-level accession mapping and the case-study statistics are provided in Supplementary Data 1. Session-level robustness and ablation scores will be deposited with the article archive. Raw sequencing files and large reference databases will not be redistributed. Their sources and releases will be documented.

Code availability

MicroBioAgent source code, domain procedures, agent instructions, analysis prompts, environment specifications, and the session-runtime extension (distributed as a version-pinned OpenClaw patch pack) are available in the project repository. Scripts used to generate the reported statistical results and figures will be included in the accompanying archive.

References

[1] Rob Knight, Alison Vrbanac, Bryn C. Taylor, et al. "Best Practices for Analysing Microbiomes". In: *Nature Reviews Microbiology* 16.7 (2018), pp. 410–422. DOI: 10.1038/s41579-018-0029-9.

[2] W. M. Tong and Yuki Chan. "GenePiper, a Graphical User Interface Tool for Microbiome Sequence Data Mining". In: *Microbiology Resource Announcements* 9.1 (2020), e01195–19. DOI: 10.1128/MRA.01195-19.

[3] Yong-Xin Liu, Yuan Qin, Tong Chen, Meiping Lu, Xubo Qian, Xiaoxuan Guo, and Yang Bai. "A Practical Guide to Amplicon and Metagenomic Analysis of Microbiome Data". In: *Protein & Cell* 12.5 (2021), pp. 315–330. DOI: 10.1007/s13238-020-00724-8.

[4] Johannes Köster and Sven Rahmann. "Snakemake—a scalable bioinformatics workflow engine". In: *Bioinformatics* 28.19 (2012), pp. 2520–2522. DOI: 10.1093/bioinformatics/bts480.

[5] Paolo Di Tommaso, Maria Chatzou, Evan W. Floden, et al. "Nextflow enables reproducible computational workflows". In: *Nature Biotechnology* 35 (2017), pp. 316–319. DOI: 10.1038/nbt.3820.

[6] Juexiao Zhou, Bin Zhang, Guowei Li, Xiuying Chen, Haoyang Li, Xiaopeng Xu, Siyuan Chen, Wenjia He, Chencheng Xu, Liwei Liu, and Xin Gao. "An AI Agent for Fully Automated Multi-Omic Analyses". In: *Advanced Science* 11 (2024), p. 2407094. DOI: 10.1002/advs.202407094.

[7] Kexin Huang, Serena Zhang, Hanchen Wang, Yuanhao Qu, Yingzhou Lu, Yusuf Roohani, et al. "Biomni: A General-Purpose Biomedical AI Agent". In: *bioRxiv* (2025). Preprint. DOI: 10.1101/2025.05.30.656746.

[8] Dechao Bu, Jingbo Sun, Kun Li, Zihao He, Wei Huang, Jinlin Hu, Shanshan Zhang, Shuangshuang Lei, Peipei Huo, Zhihao Wang, et al. "Empowering AI Data Scientists Using a Multi-Agent LLM Framework with Self-Evolving Capabilities for Autonomous, Tool-Aware Biomedical Data Analyses". In: *Nature Biomedical Engineering* (2026). DOI: 10.1038/s41551-026-01634-6.

[9] Houcheng Su, Junning Feng, Yawen Lu, Yucheng Xu, Jinming Yang, Haojie Lu, Jixin Yang, Xu Yang, Sirui Xie, Weicai Long, Chengrui Wang, Yusen Hou, Tingyu Zhu, and Yanlin Zhang. "BioMaster: Multi-Agent System for Automated Bioinformatics Analysis Workflow". In: *Patterns* 7.8 (2026), p. 101611. DOI: 10.1016/j.patter.2026.101611.

[10] Shreya Johri, Erica Pimenta, Josephine Yates, Jingxin Fu, Erik L. Bao, Hyeji Jun, Brendan Reardon, Sasha Bacot, Maha Shady, Doris Fu, Wenbin Mei, Sabrina Y. Camp, Jihye Park, and Eliezer Van Allen. "Evaluating Agentic AI for Biological Discovery in Autonomous and Copilot Settings". In: *bioRxiv* (2026). Preprint. DOI: 10.64898/2026.06.04.729919.

[11] Haohong Zhang, Zuqi Li, Philip Naderev Panuringan Lagniton, Zikang Wang, Linfeng Zhao, Wenxuan Li, Pengmin Duan, Xiaosen Jiang, and Kang Ning. "MetaClaw: An Auditable AI Agent for End-to-End, Multi-Directional Metagenomic and Multi-Omics Analysis". In: *bioRxiv* (2026). Preprint, version 2, posted 3 September 2026. DOI: 10.64898/2026.07.21.739769.

[12] Jiakang Yin, Ziliang Zhang, Chengcheng Zhu, Taotao Wang, Ruihong Wang, and Lifang Ruan. "Heritability of Tomato Rhizobacteria Resistant to Ralstonia solanacearum". In: *Microbiome* 10 (2022), p. 227. DOI: 10.1186/s40168-022-01413-w.

[13] Jun Yu, Qiang Feng, Sunny Hei Wong, Dongya Zhang, Qiao Yi Liang, Youwen Qin, Longqing Tang, Hui Zhao, Jan Stenvang, Yanli Li, et al. "Metagenomic Analysis of Faecal Microbiome as a Tool towards Targeted Non-Invasive Biomarkers for Colorectal Cancer". In: *Gut* 66.1 (2017), pp. 70–78. DOI: 10.1136/gutjnl-2015-309800.

[14] Ronja Marlonsdotter Sandholm, Gordon Jacob Boehlich, Ørjan Dahl, Ravindra R. Chowreddy, Anton Stepnov, Gustav Vaaje-Kolstad, and Sabina Leanti La Rosa. "Microbial Degradation of a Widely Used Model Polyethylene Is Restricted to Medium- and Long-Chain Alkanes and Their Oxidized Derivatives". In: *The ISME Journal* 19.1 (2025), wraf276. DOI: 10.1093/ismejo/wraf276.

[15] Nikita Mehandru, Amanda K. Hall, Olesya Melnichenko, et al. "BioAgents: Bridging the Gap in Bioinformatics Analysis with Multi-Agent Systems". In: *Scientific Reports* 15 (2025), p. 39036. DOI: 10.1038/s41598-025-25919-z.

[16] Qi Xin, Quyu Kong, Hongyi Ji, Yue Shen, Yuqi Liu, Yan Sun, Zhilin Zhang, Zhaorong Li, Xunlong Xia, Bing Deng, and Yinqi Bai. "BioInformatics Agent (BIA): Unleashing the Power of Large Language Models to Reshape Bioinformatics Workflow". In: *bioRxiv* (2024). Preprint. DOI: 10.1101/2024.05.22.595240.

[17] Sebastian Lobentanzer, Shaohong Feng, Noah Bruderer, Andreas Maier, BioChatter Consortium, Cankun Wang, Jan Baumbach, Jorge Abreu-Vicente, Nils Krehl, Qin Ma, Thomas Lemberger, and Julio Saez-Rodriguez. "A Platform for the Biomedical Application of Large Language Models". In: *Nature Biotechnology* 43.2 (2025), pp. 166–169. DOI: 10.1038/s41587-024-02534-3.

Supplementary Information

Case-study accessions and statistics are provided in Supplementary Data 1.

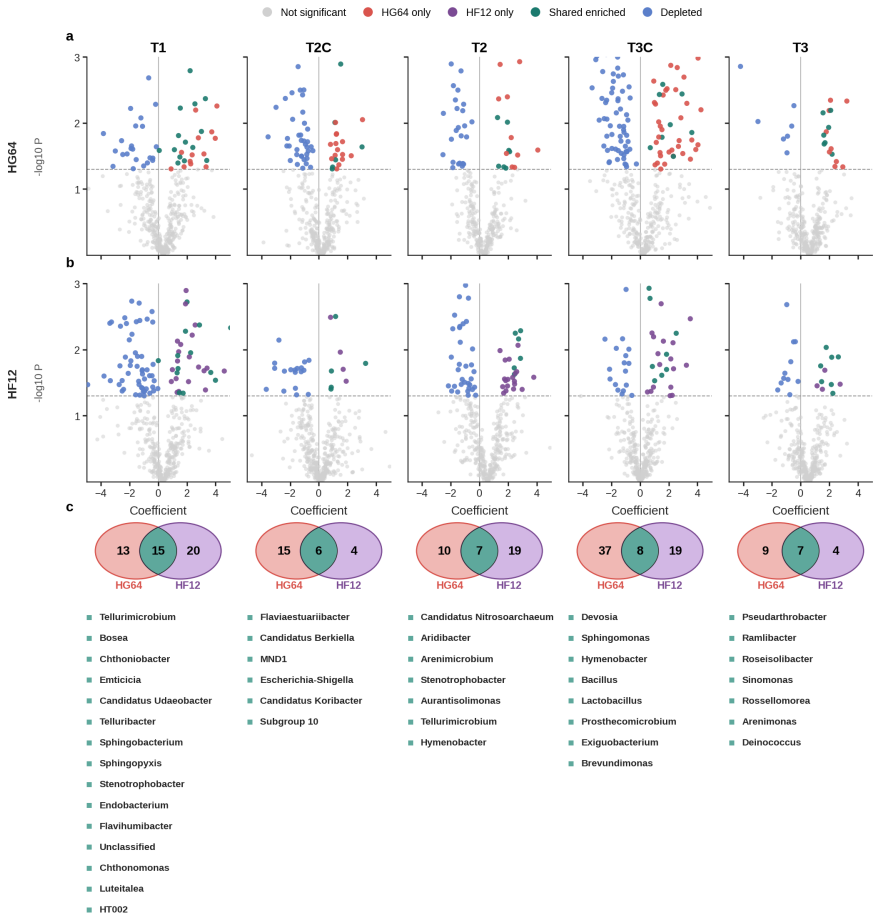


Figure S1: Differentially enriched genera in the resistant tomato parent HG64 and progeny HF12. MaAsLin 3 compared each resistant cultivar with pooled susceptible cultivars at each sampling point. Shared enrichment denotes a positive coefficient and $P < 0.05$ in both contrasts.

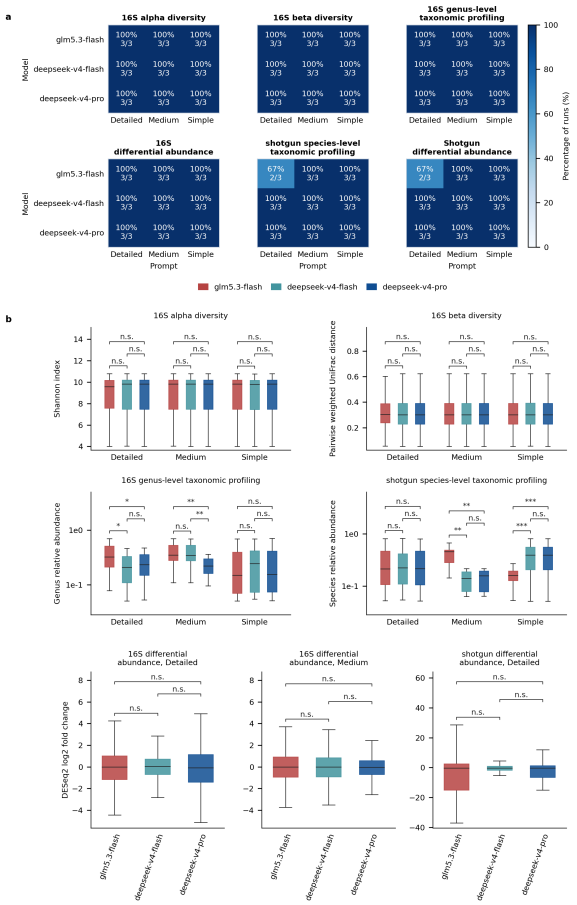


Figure S2: Overall task completion and differences in analytical outputs across models within each prompt level. (a) Completion of six requested analyses. Each cell is the number of sessions, out of three, in which the analysis was completed. The `glm5.3-flash` detailed shotgun cells are 2/3 because one session did not analyse the shotgun libraries. (b) Shannon index, pairwise weighted UniFrac distance, relative abundance and DESeq2 $\log_2$ fold change compared among models within each prompt. Shannon points are session means of per-sample indexes. UniFrac points are session means of distances between unordered sample pairs ($n = 1{,}770$). Relative-abundance points are session means for sample and taxon entries whose mean was at least 5% in every compared group. DESeq2 points are means across sessions that ran DESeq2, for taxon, cultivar and timepoint entries shared by the compared models (Supplementary Table S4, panel a); the medium shotgun prompt had no such sessions. Brackets are two-sided Mann–Whitney U tests with Holm correction within each set of three pairs. n.s., Holm-adjusted $P \geq 0.05$; *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$. Samples, sample pairs, taxa and contrasts come from one experiment and are not independent replicates.

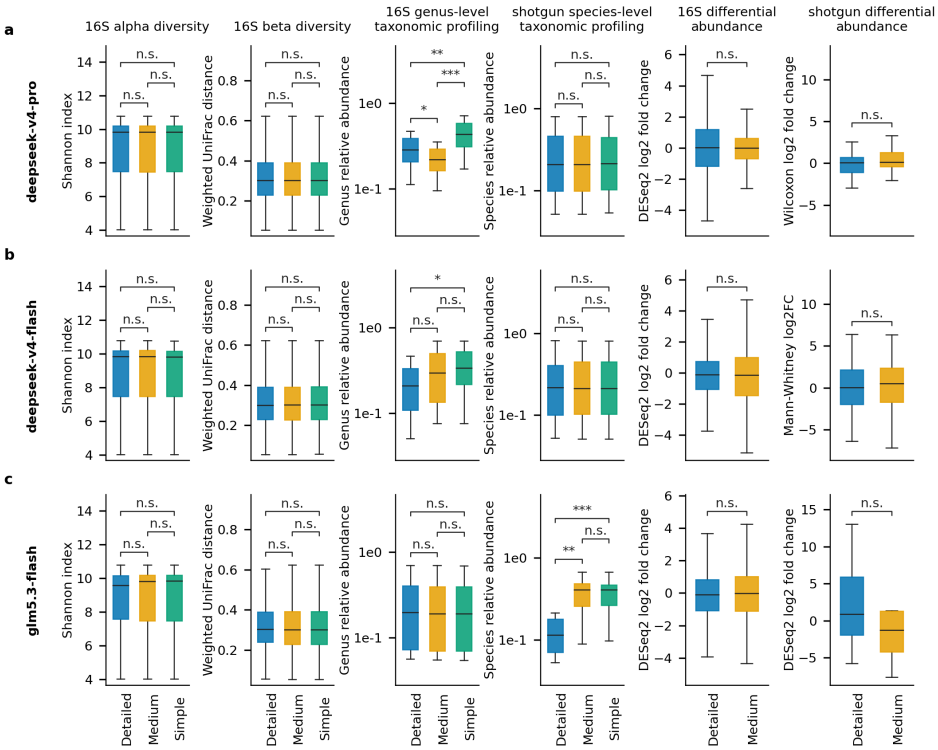


Figure S3: Differences in analytical outputs across prompt levels within each model. Rows are `deepseek-v4-pro`, `deepseek-v4-flash` and `glm5.3-flash`. The first four columns follow the point definitions in Supplementary Figure S2. The differential-abundance columns compare the detailed and medium prompts only. 16S values are DESeq2 $\log_2$ fold changes. Shotgun values are Wilcoxon, Mann–Whitney U and DESeq2 $\log_2$ fold changes for `deepseek-v4-pro`, `deepseek-v4-flash` and `glm5.3-flash`, respectively (Supplementary Table S4, panel b), and are not compared across rows. Brackets in the first four columns are two-sided Mann–Whitney U tests with Holm correction within each set of three prompt pairs. Each differential-abundance bracket is a single pair. n.s., adjusted $P \geq 0.05$; *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$. Samples, sample pairs, taxa and contrasts come from one experiment and are not independent replicates.

Table S1: Published datasets, sequencing-library counts, analysis tasks, and evaluation targets for the three MicroBioAgent case studies.

Case	BioProject	Sequencing libraries	Analysis route	Primary evaluation target
Tomato rhizosphere	PRJNA777724	60 16S; 36 matched shotgun	16S diversity, taxonomy, differential abundance; MetaPhlAn profiling	Recovery of inoculation- and resistance-associated community signals
CRC faecal metagenomes	PRJEB10878	128 shotgun	MetaPhlAn3, HUMAnN3, differential analysis, nested-CV classification	Taxonomic and functional case–control analysis from reads
Enrichment metagenomes	PRJEB97175	6 shotgun	MAG construction, dereplication, taxonomic assignment, abundance, eggNOG annotation	Genome-resolved taxonomic and functional characterization

Notes: Sample-level run accessions are listed in Supplementary Data 1.

Table S2: Software and reference versions recorded for the case-study analyses.

Software or database	Version or release	Conda environment	Case
MetaPhlAn	3; `mpa_v30_CHOCOPhlAn_201901`	`biobakery3`	Cases 1 and 2
HUMAnN	3; build not recorded	—	Case 2
MaAsLin3	1.2.0	—	Case 1
SILVA	138.2, V3–V4	—	Case 1
CheckM2	1.1.0	—	Case 3
CoverM	0.7.0	—	Case 3
GTDB-Tk	2.7.2; database R232	—	Case 3
eggNOG-mapper	2.1.13; eggNOG 5.0	`eggnog_mapper`	Case 3

Notes: Versions are those written in the archived analysis outputs. An em dash means the conda environment was not recorded. MetaPhlAn2 profiles were not present in the archive, so no MetaPhlAn2 database release is listed. The MetaPhlAn row is the MetaPhlAn3 run recorded in the profile headers.

Table S3: Reported capabilities of bioinformatics agents relevant to end-to-end microbiome analysis. The three microbiome columns specifically concern analysis beginning with sequencing reads.

	Agent operation		Microbiome analysis from reads			Deployment
Agent	Autonomous workflow execution	Human-in-the-loop	16S amplicon	Reference-based shotgun	Assembly / MAG	On-premises analysis
BIA	✓	—	—	—	—	✓
BioChatter	—	✓	—	—	—	✓
AutoBA	✓	—	—	—	—	✓
Biomni (open-source)	✓	—	—	—	—	✓
BioAgents	—	—	—	—	—	—
M3A	✓	✓	—	—	—	✓
BioMedAgent	✓	—	—	—	—	✓
BioMaster	✓	—	—	✓	—	✓
MetaClaw	✓	—	✓	✓	✓	✓(partial)
MicroBioAgent	✓	✓	✓	✓	✓	✓

Notes: ✓, reported support, —, support not established from the reviewed sources, which does not demonstrate inability. The comparison draws on the published descriptions of the related systems [6–11, 15–17]. Biomni denotes the open-source release. MetaClaw reports assembly and genome-resolved binning through FlowHub [11]. Its partial on-premises designation reflects the use of FlowHub for upstream metagenomic processing and local containers for downstream analysis. This scope comparison is not a performance benchmark.

Table S4: Robustness comparisons of differential-abundance outputs. (a) DESeq2 sessions included in the cross-model comparison of differential-abundance effect sizes. Each row is one model. Session counts are the replicates, out of three, whose DESeq2 result entered the comparison. The medium shotgun prompt produced no DESeq2 results. (b) Differential-abundance methods and replicates averaged in the detailed-versus-medium prompt comparison. Each row is one comparison. Session counts are the replicates, out of three, whose result for that method entered the average.

(a)

Model	Detailed sessions	Medium sessions	Detailed entries shared	Medium entries shared
16S rRNA gene, genus				
`deepseek-v4-pro`	1/3	1/3	1,724	2,659
`deepseek-v4-flash`	2/3	2/3	1,724	2,659
`glm5.3-flash`	3/3	1/3	1,724	2,659
Shotgun metagenome, species				
`deepseek-v4-pro`	1/3	—	228	—
`deepseek-v4-flash`	2/3	—	228	—
`glm5.3-flash`	2/3	—	228	—

Notes: Each entry represents the DESeq2 $\log_2$ fold change of a taxon for a given resistant cultivar (HG64 or HF12) and sampling timepoint relative to the susceptible control (averaged across susceptible controls within a session when A57 and HG70 were reported separately). Values were averaged across the included sessions without imputing missing entries as zero. An entry was shared when it was reported across all three models, so the shared count is the same in each row of a block. One `glm5.3-flash` detailed replicate did not analyse shotgun libraries.

(b)

Model	Method	Detailed sessions	Medium sessions	Entries compared
16S rRNA gene, genus				
`deepseek-v4-pro`	DESeq2	1/3	1/3	891
`deepseek-v4-flash`	DESeq2	2/3	2/3	5,108
`glm5.3-flash`	DESeq2	3/3	1/3	3,984
Shotgun metagenome, species				
`deepseek-v4-pro`	Wilcoxon	1/3	1/3	39
`deepseek-v4-flash`	Mann–Whitney U	1/3	1/3	120
`glm5.3-flash`	DESeq2	2/3	1/3	52

Notes: The simple prompt was not included. Each entry represents the $\log_2$ fold change of a taxon for a given resistant cultivar (HG64 or HF12) and sampling timepoint relative to the susceptible control (averaged across susceptible controls within a session when A57 and HG70 were reported separately). Values were averaged across the included replicates without imputing missing entries as zero, and entries reported under both prompts were compared. DESeq2 values are negative-binomial $\log_2$ fold changes. Wilcoxon and Mann–Whitney U values are $\log_2$ ratios of group means reported with the rank test, not the rank statistic. One `deepseek-v4-flash` detailed replicate reported only ASV-level DESeq2 results and was excluded from the genus comparison. One `glm5.3-flash` detailed replicate did not analyse shotgun libraries.